

Exp -12 Decision Tree Classifier

Program:

```
from sklearn.tree import DecisionTreeClassifier

X = [[25,30000],[35,60000],[45,80000],[20,20000],[50,90000]]
y = [0,1,1,0,1]

clf = DecisionTreeClassifier()

clf.fit(X, y)

test = [[40,70000]]

result = clf.predict(test)

print("Prediction:", result)

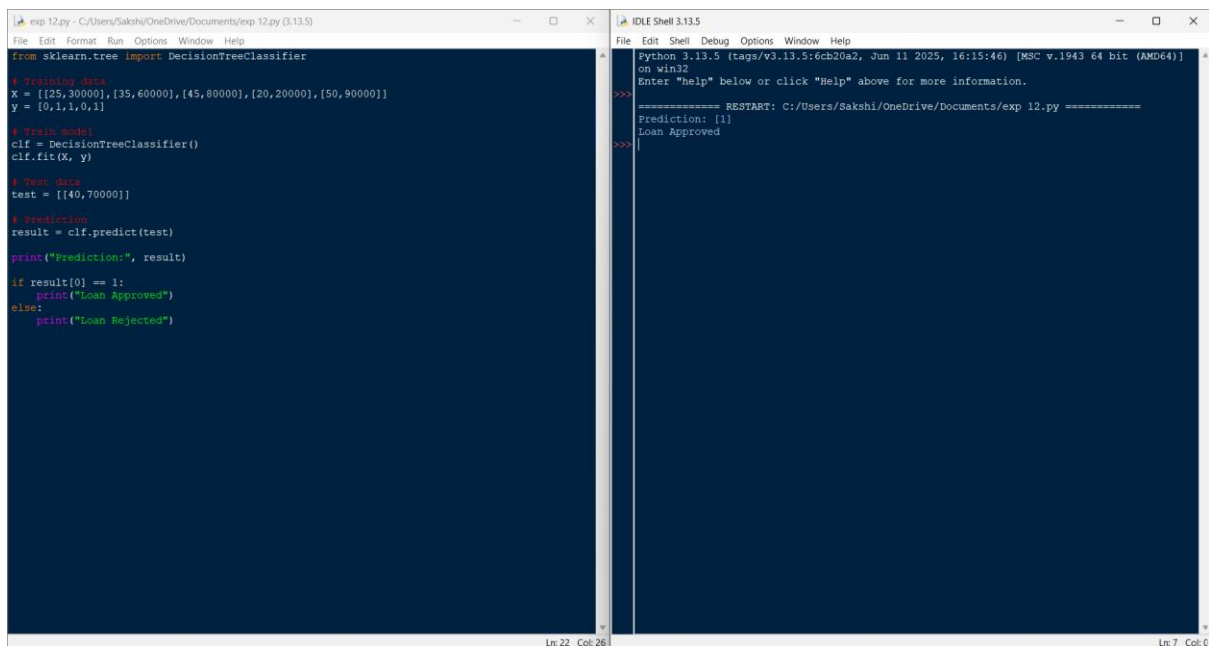
if result[0] == 1:

    print("Loan Approved")

else:

    print("Loan Rejected")
```

Output:



The screenshot displays a Python IDE with two windows. The left window, titled 'exp 12.py', contains the code for a Decision Tree Classifier. The right window, titled 'IDLE Shell 3.13.5', shows the execution output. The code defines training data X and target labels y, trains a DecisionTreeClassifier, and uses it to predict the class for a test sample [40, 70000]. The output shows the prediction as 1, indicating 'Loan Approved'.

```
exp 12.py - C:/Users/Sakshi/OneDrive/Documents/exp 12.py (3.13.5)
File Edit Format Run Options Window Help
from sklearn.tree import DecisionTreeClassifier

# Training data
x = [[25,30000],[35,60000],[45,80000],[20,20000],[50,90000]]
y = [0,1,1,0,1]

# Train model
clf = DecisionTreeClassifier()
clf.fit(X, y)

# Test data
test = [[40,70000]]

# Prediction
result = clf.predict(test)

print("Prediction:", result)

if result[0] == 1:
    print("Loan Approved")
else:
    print("Loan Rejected")

Ln: 22 Col: 26

IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6ch20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/Sakshi/OneDrive/Documents/exp 12.py =====
Predictions: [1]
Loan Approved
>>>

Ln: 7 Col: 0
```